SLIIT | FACULTY OF COMPUTING

BUDDY

Affordable, Emotionally Intelligent Robotic Companion & AI Security Assistant

Final Research Presentation · R26-SE-040

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SLIIT | Faculty of Computing

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PP2 feedback → targeted development

Each panel observation, what changed, and what is still pending live validation.

AREA	PP2 OBSERVATION	ACTION AND EVIDENCE
Overall demo	Demonstration judged not sufficient for the group.	Rebuilt around a live Pi + ROS 2 + Flutter demo. Camera, object and proximity evidence captured (slides 9 to 14).
Emotion → response	Emotion-only work questioned; research and actions to improve the user's mood expected.	Expanded vision, pose/fall and emotion-responsive dialogue/music design. Live music-trigger validation pending.
Navigation component	Component not demonstrated at PP2.	ROS 2 obstacle safety plus the Yohan_Mapping branch. Live proximity UI working; autonomous room mapping not verified.

Problem in brief

Companionship

Accessible interactive assistance for everyday needs, especially when human availability is limited.

Safety awareness

Detect obstacles, unknown visitors and possible falls, with visibility through a phone app.

Affordability

Practical AI and robotics capabilities built around a Raspberry Pi and modular sensors.

Research objective: an integrated, low-cost prototype that combines offline-first interaction, computer vision and robot safety awareness.

Design excellence and system differentiation

1

Multimodal companion

Offline voice, optional local LLM, camera perception and a mobile interface.

2

Privacy-oriented identity

Trained identity embeddings and an explicit enrollment flow.

3

Modular ROS 2 control

Ultrasonic monitoring and motor safety kept separate from vision.

4

Teach-by-example vision

Objects learned from samples, plus a live, non-motorized 2D proximity scan.



Physical BUDDY prototype: evidence photo

Full-stack architecture

SENSING

NoIR camera

HC-SR04 × 4

Microphone

›

ON-DEVICE INTELLIGENCE

ROS 2 Jazzy

Vision inference

Vosk · optional Ollama

›

API + MOBILE

Flask MJPEG / JSON endpoints

Flutter dashboard

›

OUTPUT

Speech · recognition

Alerts · object teaching

Proximity visualization

Vision test API is kept separate from motor commands, so camera experiments cannot drive the robot.

Individual component: Ashandth

AU

Ashandth Uthayashankar

IT22272522 · Offline voice, ROS 2, integration

Speech + reasoning

Wake word "Buddy", Vosk/Whisper-oriented STT, intent responses and offline-first TTS. Optional local Ollama/Llama path.

ROS 2 + hardware

Raspberry Pi 4B, Jazzy integration, sensor/motor/neck coordination, vision test API kept apart from motor commands.

20 Oct proof

Pi NoIR camera streaming, known-face recognition, 5 learned objects / 30 samples, live ultrasonic Flask → Flutter view.

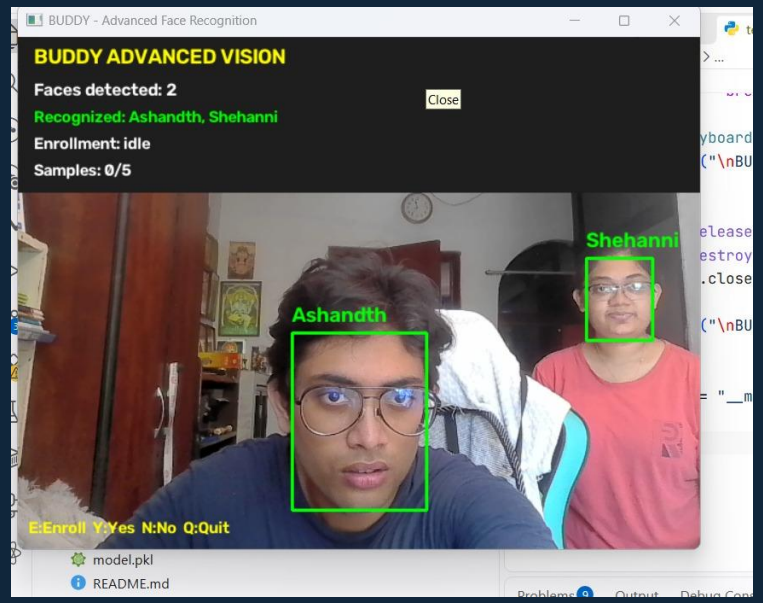
Individual component: Sobiya

SS

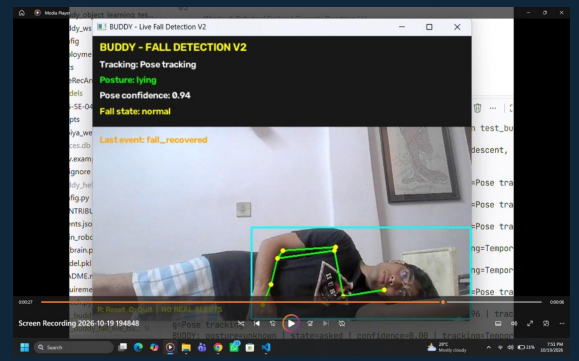
Sobiya Anton Suresh
IT22203694

PP2 baseline

Facial emotion recognition and mood estimation. The panel asked for more than labelling: the robot should respond to mood.



Face identification: desktop demo



Pose and fall tracking: recorded demo

Extended after feedback

Identity, unknown-person detection, object vision, pose/fall and gesture work. Mood → supportive speech/music proposed; live validation pending.

Individual component: SSY / Yohan

YW

SSY Wickramasinghe (Yohan)

IT21816086 · ROS 2 navigation, ultrasonic mapping

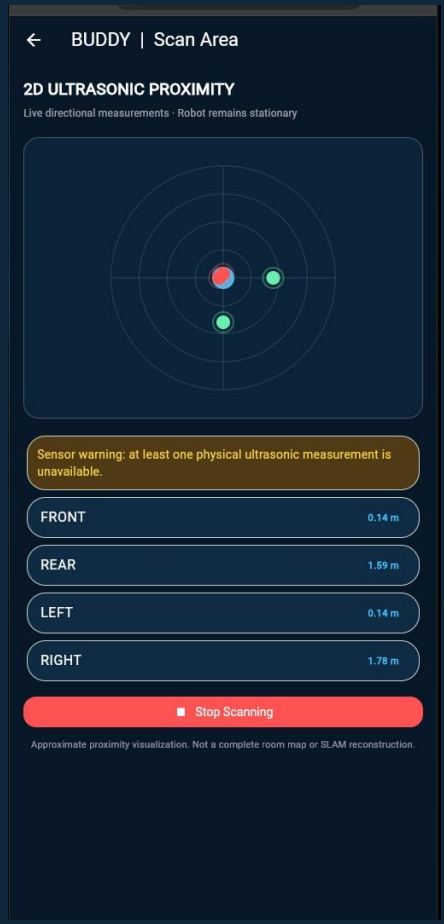
DATA PATH, VERIFIED ON THE PI, 20 OCT

4× HC-SR04	front · rear · left · right
/obstacle_status	ROS 2 topic
Mapping subscriber	read-only, no motor access
Flask	/api/mapping/status
Flutter	Scan Area plot

Status

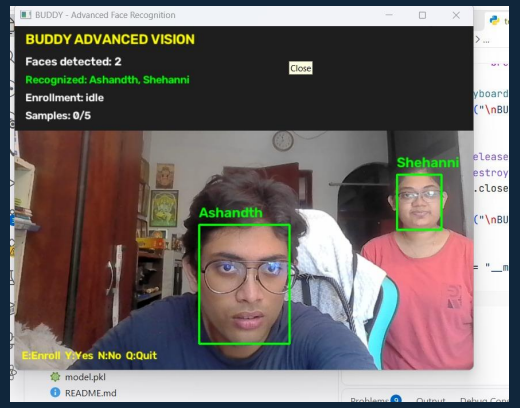
- **Live proximity: verified**
- **Room-scale SLAM: no**
- **Degraded flag: still true**

Four-direction ranging, not a physical room map.

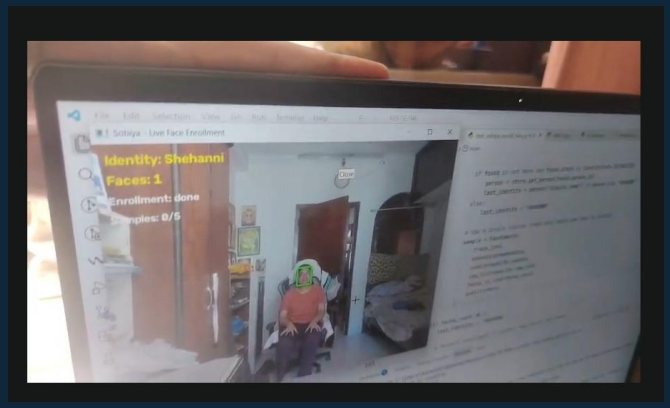


Live proximity scan, 20 Oct

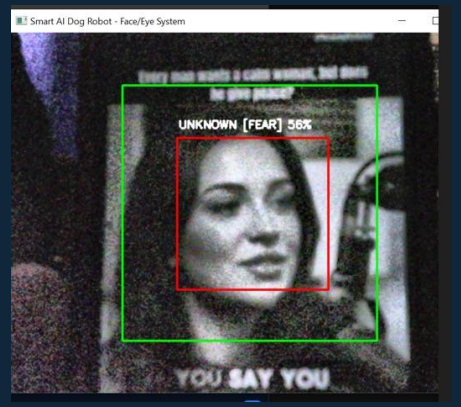
Face identity and people-aware security



Known-person recognition



Distance-range test

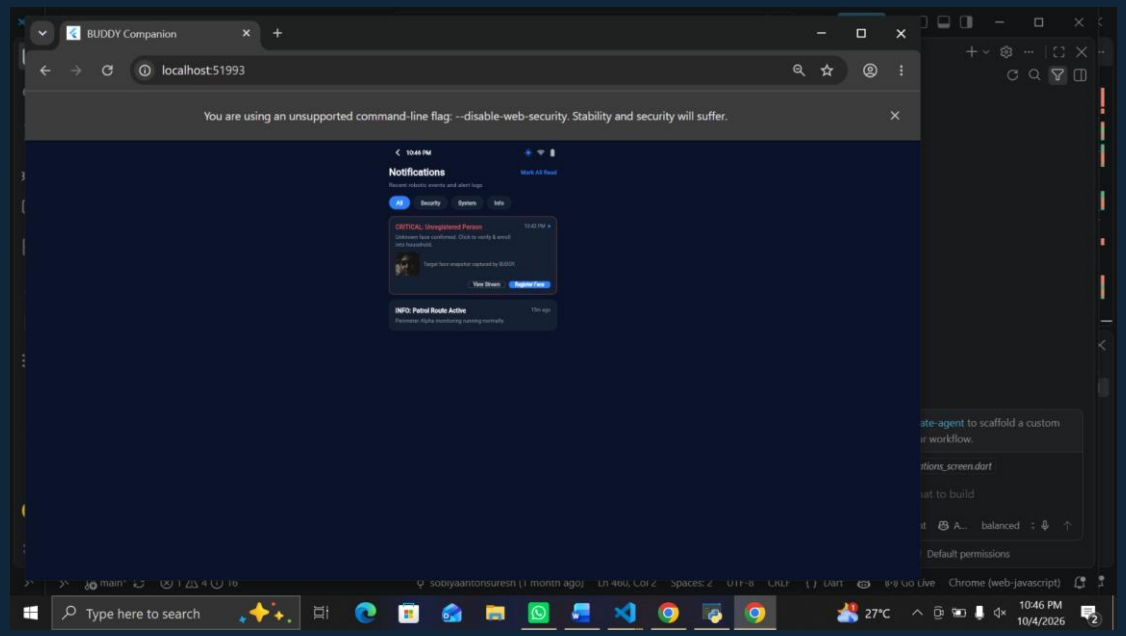


Unknown person

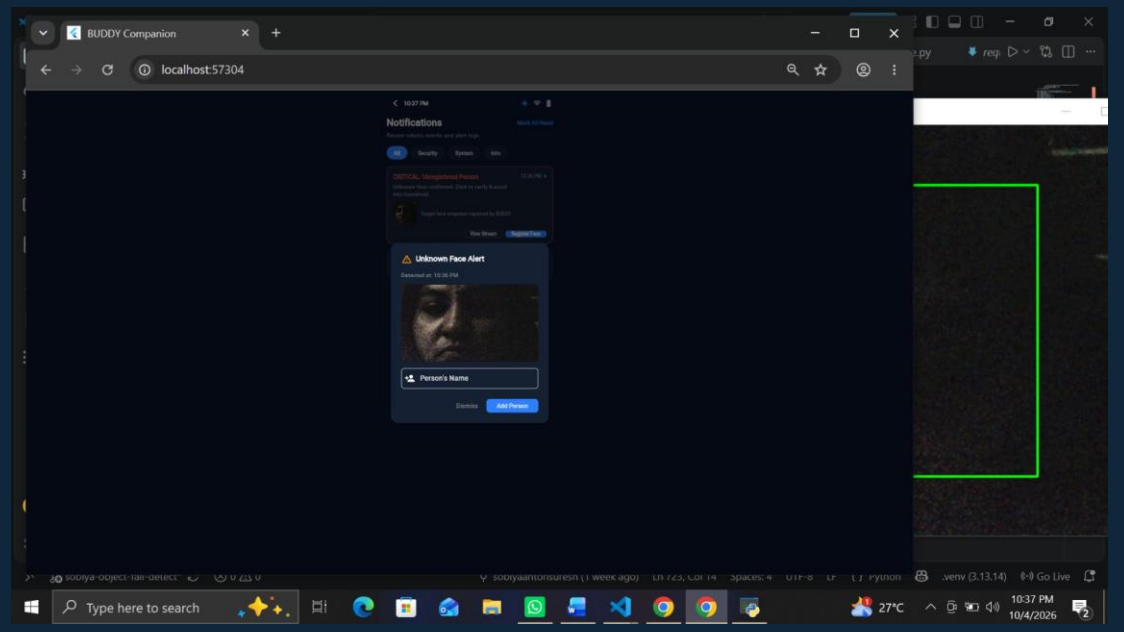


Mood and identity display (app)

Proactive monitoring: detection → alert



Unknown-person alert / enrollment workflow



Manual enrollment screen

Live Pi verification: "Ashandth" registered with 10 face embeddings. Other profiles require authorized transfer.

Object learning: teach once, recall later



Desktop proof: bounding box and predicted label

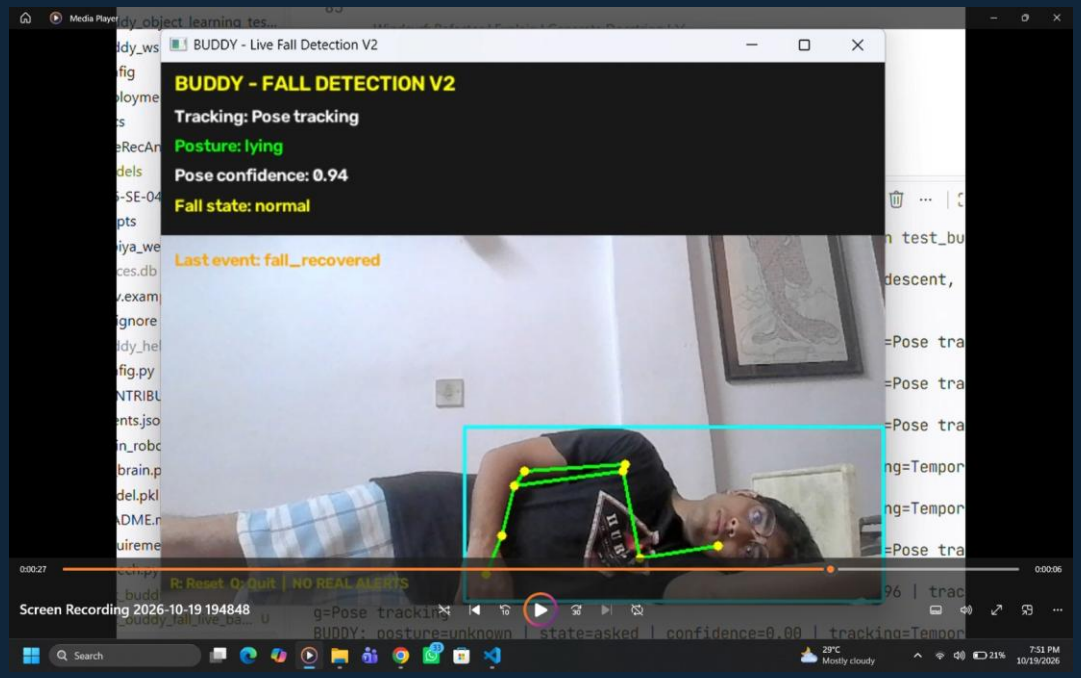
Cross-device transfer

Five learned items imported to the Raspberry Pi: blue box, purse, pen, mike box and phone (6 samples each).

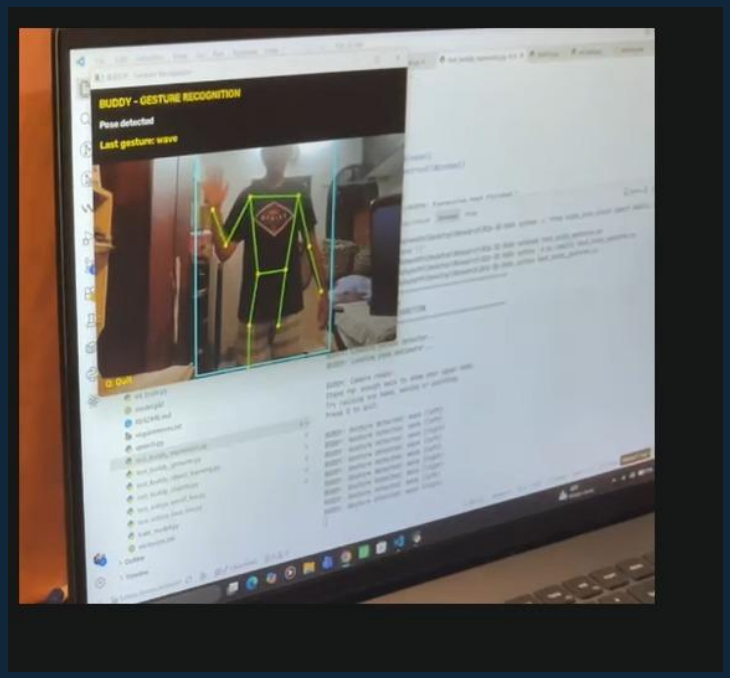
Live inference on the Pi

"mike box" at 70.44% and "pen" at 75.67% similarity. Both fell below the confidence threshold, so the classifier returned "uncertain".

Human activity, falls and gesture exploration



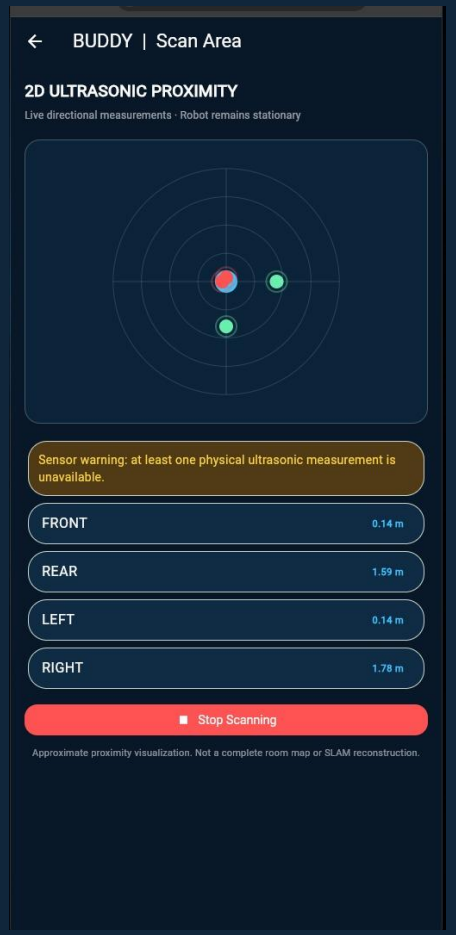
Pose: lying, confidence 0.94, fall state normal (recorded Windows demo)



Hand-pose visualization

The Pi API exposes pose processing, but live skeleton, posture and fall overlays in Flutter need more backend data. Recorded proof and Pi integration are separate validation stages.

Raspberry Pi + Flutter: live demo



Live radar, 4 sensors

Validated now

ROS 2 /obstacle_status → read-only mapping subscriber → Flask /api/mapping/status → Flutter Scan Area.

Vision and mobile

NoIR camera live feed, object-learning menu, person identity store and dedicated AI Tests screens.

Demonstrated achievements and evaluation

5	30	10	4
Learned objects	Object samples	Ashandth face embeddings	Ultrasonic directions

- Live object recognition on the Pi: mike box 0.7044, pen 0.7567 (both reported "uncertain").
- ROS mapping endpoint returned HTTP 200; camera connected and vision ready in tested sessions.
- Screenshots demonstrate desktop pose/fall, identity and alert workflows.

Limitations, integrity and next steps

Current limitations

Four-direction ultrasound is not full SLAM, and the degraded-sensor flag is still true. Live Flutter pose/skeleton overlays and emotion-triggered speech or music are not yet validated end-to-end.

Immediate next milestones

Test emotion → supportive speech → mood-matched local audio; validate consent, false positives and opt-out. Then integrate fall/gesture overlays and benchmark accuracy.

Priority: preserve physical motor-safety behaviour, and claim benchmark accuracy only after a reproducible evaluation.

BUDDY: demonstrable impact

A unified companion prototype that listens, observes, learns and reports nearby safety conditions.

VOICE + ROS 2

Offline-first control and modular robot integration

VISION

Known faces, objects and proactive safety demonstrations

MOBILE

Camera testing and real-time ultrasonic proximity visualization

Thank you · Questions and live demonstration

APPENDIX · TRACEABILITY

Evidence and source audit

- Structure follows the SLIIT Final Presentation template.
- Visual evidence: 12 screenshots and photos captured during testing.
- GitHub branches verified: Sobiya, Sobiya_AdvancedVision, sobiya-object-fall-detect, Yohan, Yohan_Mapping, buddy-ros2, integration/buddy-flutter-api.
- Live Raspberry Pi ROS / Flask / Flutter console results captured on 20 Oct 2026.
- A branch existing does not by itself establish physical-world deployment; the slides keep the two separate.